ActInf Livestream #017.0 ~ Information flow in context-dependent hierarchical Bayesian inference

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hello and welcome everyone to the active inference lab and to the active inference live stream we're here in active inference live stream 17.0 and it's march 3rd 2021 i'm daniel and i'm here with two of my colleagues who can introduce themselves blue my name is sarah davis i'm a person who's interested in things i'm a student in leibniz university in a master's program in philosophy of science and i've been an engineer and a bunch of other things so i'm just generally interested in how it's all connected i'm blue knight i'm an independent research consultant based out of new mexico cool well we're here for a fun discussion so thanks both for joining at the active inference lab we're an experiment in online team communication learning and practice related to active inference you can find us at our links and socials this is a recorded and an archived and a hastily produced live stream so please provide us with feedback so that we can be improving on our work all perspectives and backgrounds are welcome here and we'll follow good live stream etiquette so here we are in paper number 17 and we've just completed our first quarterly active inference lab round table so check that out on our channels if you haven't and today we're going to be setting the context for 17.1 and 17.2 the paper that we're discussing in 17 is information flow in context dependent hierarchical bayesian inference by chris fields and james f glazebrook it was published in october and what's funny is that the dot zero videos have always been about context set in context and we even wrote in the previous videos the video's introduction and its context it's not a review or the final word and then it got really meta with this paper because it's a paper about context the punchline or a punchline of this paper is that we can integrate various approaches from mathematics mostly to achieve a scale-free formulism that leads to an interpretation of nested systems where communicating systems are engaged in active inference and distinguished by informational or statistical patterns or they're engaged in something that's related to active inference let's say because it's pretty out there with what it is and it's a challenging but also a fun paper so we're gonna start with the goals and the claims and the abstract and at that point it might be like wait what with the roadmap headers or with abstract claims it's like driving through a country where you don't speak the language perhaps and that's okay because the bulk of this video in the keywords are going to be going through and kind of unpacking a lot of the key terms that are gonna matter for even just understanding the lay of the land and then we've pulled out a few other key topics some quotations didn't go as much into the formalisms of the paper itself but we're looking forward to doing that with the authors in 17.1 and so we're just looking forward to that in 17.1.2 we'll be discussing the same paper so read it and save and submit your questions and get in touch if you want to participate and also we're looking forward to your chats uh during this presentation giving us a little support along the way so here we go with the goals and claims of the paper under discussion so here are um the goals and the claims so maybe blue do you want to read the author's goals and claims sure so the goals in the paper are twofold first to model intrinsic or true contextuality using the general category theoretic methods of two spaces and channel theory and second to employ this formulation to reconstruct hierarchical bayesian inference in a context-dependent way it goes on to say here we have formulated an approach to intrinsic contextuality in general category theoretic terms a set of observations exhibits intrinsic if no cocoon can be constructed over the observables that produce them oh cone cocoon who knows how to say it um so they're doing a couple of things here they have two main goals so it's great that they're laying it out clearly the first one is about context and they want to be having a more advanced way to talk about context and it turns out that they're going to be doing that using category theory and two spaces and channel theory all of which we're going to be going into at least a little bit and then secondly they're going to be linking all of this stuff that they just did up here with context choose channel theory category theory and they're combined combining that with context dependent bayesian inference we've talked a lot about active inference and bayesian inference and so this is sort of the hinge where we connect some of these ideas to active inference and then this is a quote from their conclusion which is that they have formulated a new way of modeling contextuality in very general category theory terms specifically relating to this co-cone construction but the category theoretic term allows them to move between a few different areas that we're going to be walking through today so here we go with the abstract i'll read the first of the three slides of the abstract and anyone can give any thoughts or just start reading the second part of the abstract recent theories developing broad notions of context and its effects on inference are becoming increasingly important in fields as diverse as cognitive psychology information science and quantum information theory and computing here we introduce a novel in general approach to the characterization of using the techniques of two spaces and channel theory viewed as general theories of information flow this involves introducing three essential components into the formalism events conditions and measurement any thoughts either of you two or someone can just start reading the second part of the abstract let's go to the second part of the abstract i think we'll have a lot of time for um off-the-cuff remarks soon so sarah do you want to go for the uh sure incorporating these factors in relationship to conditional probabilities leads to information flows both in the setting of two spaces and the latter provides a representation of semantic content using local logistics from which conditionals can be derived we employ these features to construct cone cacone diagrams commutativity commutativity of which enforces inferential coherence um with these built a scale-free architecture incorporating a bayesian-like hierarchical structure in which there is an interpretation of active inference in markov blankets nice and then blue how about finish out the abstract we compare this architecture with other theories of contextuality which we briefly review we also show that this development of ideas conveniently accommodates negative probabilities leading to the notion of signed information flow and address how quantum can be interpreted within this model finally we relate contextuality to the frame problem another way of characterizing a fundamental limitation on the observational and information inferential capabilities of finite agents perfect so that is the abstract that's how the authors wanted to present and communicate their work to a broad audience of specialists and non-specialists and again we're going to be going into basically all the words that might be non-obvious in this abstract and the keywords that come into play that's what we wanted to be unpacking today so we're gonna look over their roadmap because just like the abstract is how they want to communicate what they did the roadmap is how they've chosen to organize what they've done and get us from a to z and then we're going to be really going into this road trip and unpacking what some of these key terms are it starts with an introduction which i guess is helpful context and then one of the focal ideas of the is this chew space or the chew construction which we're going to be talking about the history of some examples some related topics and just right off the bat when we saw that examples of two spaces and chew morphisms included topological spaces so that's all of network theory and graphs and things like that so that's one big area but we're going to be talking about two spaces as something that connects all of network theory and topological space to types as processes probability spaces bayesian belief networks event space structure shoe flows kovac libra divergence and the free energy so we're going to be popping up a level from things that are already pretty abstract but it's going to be done in a way that's quite useful especially for formulating and it's actually really interesting because a lot of times people will say that these um topics in the examples that they're general but they actually don't have enough context for example you can make a social network but that wouldn't be the total context of the world if you only have some of the users on the social network so maybe there's a way where we're going to be able to step up from some of our previous conceptions and introduce context potentially by default this leads the authors to a discussion of probability and how to use probabilistic models for contextuality for modeling context and this comes to a recently derived framework called contextuality by default which will be a topic we dive into section 4 goes into some issues related to measurements and kind of the before during and after of measurements because you got to set things up and calibrate and then you got to make the measurement at a certain moment and then you have to analyze it in the context of a certain system so it's like before during and after the measurements in section 5 they head into channel theory and information channels and this was really fascinating because it's quite distinct from shannon information theory in some ways that people might not expect and i had never heard this discussed alongside shannon information theory it's like here's information theory and it's good at abc but it's not good in these three things that's where the discussion ended i didn't know that there was another area that kind of picked up where shannon left off in a sense and that's related to this idea of an infomorphism as well as classifiers conditional probabilities and this cocon diagram thinking about information channels and flows of information leads to this formulation of context compliant so modeling or following hierarchical multi-level multi-scale bayesian networks and first this section has a lot of technical detail but it involves making a connection between bayes networks approaches to probabilistic modeling and statistics and inference with context this leads to a sort of conclusion or a climax section where the hierarchical bayesian networks that were just introduced in six are brought into the context of two specific uh practices which is number one the idea of active inference i highlighted it because that's kind of our lab that's what we're here to be thinking about and then two was the frame problem which is something that's also really interesting to think about it's going to be the key motivator so although they put it um last with active inference in the frame problem we're actually going to have those topics closer to the front of the presentation so people who are listening to this from the active inference perspective or from more generally the frame problem perspective they're going to understand what motivated us to be studying the paper the way that we're doing it and hopefully what motivated chris to select this paper when he suggested it to be discussed during his visit that's the sort of uh naked road map that's the turn left at this part and turn right at this part but there was a a narrative road map that the authors actually put out and their own words are on the right so you can see that in the paper or here but on the left i just thought of this is like the semantic road map as far as the flow of the paper and i just extracted this semantic flow from the syntax that they wrote not super knowing what all these topics are but just re-phrasing what they had put previously these authors had been researching quantum and thermodynamics of separable systems in this paper they started with a recap of two spaces and choomorphisms which suggests that it's a foundational topic for the paper they then reviewed contextuality paradigms and the diversity of approaches that model they then took a formal uh diversion into this idea into various ideas of probability theory above my detailed understanding but they took an extension in that direction they then turned to semantically richer methods of channel and they're using channel theory as an even more general approach to inference rather than the syntactic information theory like shannon information theory which we'll get to they wanted something a little bigger thinking about these kinds of uh local logics of information channels leads to some new ways that they also combined ideas and then they visualize this slash represent it as this cone cocon diagram as they say as a natural scale representation of bayesian inference so scale free just means that it doesn't have an a priori scale it's not in inches or in kilometers it's just at the level of abstractness and then they have the claim that is proven that the failure of the cccd the cone cocone diagram to diagrammatically commute is a condition of the measurement operator's non-co-deployability so it's just sort of like if you have two shapes and then they fit together what they represent fits together and if you have two shapes and the puzzle pieces don't fit together there's something mathematical that the shapes represent isn't fitting together in the way that it would as if these two shapes fit together i don't know what non-co-deployability is but it's like a puzzle piece if it were co-deployable and then the part that i think will be of special interest to a lot of our lab and colleagues is they're gonna recover this new theoretical and inferential role for the markov blanket it's the locus at which free energy is minimized part one and part two the epistemic barrier that keeps contextual information hidden from the observer so when i saw that i was like whoa it sounds fascinating because it's two distinct and critical roles of a blanket and then i thought so contextual information is hidden from the observer because a lot of times we might think about context as being like what's around us but it we'll see if that's actually the way that context is being used can i jump in for a second daniel yep so i think that i the way that i understood um co-deployability like measurement operator code deployability like you can't at the same time measure position and velocity that was like one example that they gave in the paper so like because those two things like you can't if you're moving you're not at a stable position right so so it's those those two measurements are not co-deployable you can't measure those two things at the same time and that's kind of what they meant like about measurement code deployability i think cool thanks for that here's to why it matters before we even jump into the keywords which will be the bulk of our journey and this is quotes from the authors while the results presented here are primarily technical so just take a breath if it's worrying you because they wrote the paper so we don't have to and it's a technical paper and that's what we're learning by doing and exploring the paper is technical and abstract but now we're looking for where we can go it's like a car is technical but it lets you get somewhere so where are we going and why will it matter these results open a path to application in a number of areas where contextuality has largely been neglected but is important in practice that sounds really important because it sounds like there's um systems that are unknown unknowns we don't even know what to model but there might be a large category of systems where we think we've modeled it well but then there's this problem where context keeps on entering the picture or something that we didn't model throws our local model off that we spent time and energy on so if we could find a way to respectfully interface with contextuality across different areas that would be really important well what are those areas one of the models that they're interested in just like fristen and a lot of active inference researchers is neuroscience and so they suggest that the current approach they're taking takes a previous model which is the global neuronal workspace model and that's actually related to what ryan smith and christopher white will be talking to us about in um number 18. they're going to make that model contextuality compliant and provide the building blocks needed to incorporate context switching and attention switching into global neuronal workspace models so it's kind of these two cool parts of the brain which is that it can spiral in and really hone in with attention but then it can also move attention so it's like side to side in and out and that sort of context and attention switching is not dealt with well by our current models which might be about attention or inattention to a stimulus but they're not actually modeling what the regime of attention is um another then there's a formal result which might matter to some people or others that contextuality can always be associated with this sort of non-community kind of related to the puzzle pieces the context we just mentioned and then um this is interesting from a more probability theoretic point of view our results further support the authors of the contextuality by default in showing that the distinction between quantum and classical probabilities lies not in any ontological difference but rather in what has been explicitly labeled so this was really mind-bending because a lot of times it's like quantum you know theory of small things where there's an observer that has an effect in the outcomes of how the system is modeled or set up and then um the sort of disinfo debate is like who's an observer or does it matter if it's a conscious deserve you're going to go down a rabbit hole because there aren't simple answers there but the system of the particle it turns out depends on how you structure the context of the experiment just as is the case for macroscopic entities so these were some questions that so so in other words is there a continuum of uh tiny things and bigger things with context as being what is distinguishing the behavior that can seem quite different across scales so it's not that electrons are magical and they can do wave particle but nothing else exists in a similar regime it's not that we're going to get the same diffraction of people walking through a doorway but then again maybe if it's set up the right way there's more similarity than not so a few opening questions before we jump into the keywords just how will we connect across formal and informal or just plain different senses of the word and then what are some of these cases where whether they said it in the paper or not where contextuality has been largely neglected but it's important in practice so even within an informal context definition what's in an area where potentially context is not taken into account so i'll start with one and then either of you could go maybe cultural context or for understanding geopolitics or individual context for an individual relationship so it does matter but when we approach those sort of individual or collective level problems as if we can ignore context again just using it informally we're not really operating in the space that the system is in we're operating in a reduced or a distorted model that we're projecting uh hey daniel i i don't know if uh this is gonna mess up your video setup but if you're able to go into present mode on your slides it would make the whole thing a lot more readable for a small screen in the live stream which you can look at and mute in the background it looks different okay um but yeah we've got a lot of small text on these slides yeah it's um i don't know it's so worth unpacking that last bullet point it's um there's just a lot there is is how so are you reading um when i read this um you know that he's saying ontological difference versus what's explicitly labeled i immediately just think well of course because you're you know you're either looking at small scale or you're looking at macro scale but you think he's saying something something a little more subtle um i would love to have a quantum expert but just speaking with our friend jason larkin he has often pointed to um these sorts of almost like classical meets quantum crossover situations where you get the classical quote behavior from when you define the full system with the quantum observer then it behaves let's just use the twin slit for example the system behaves a certain way when you define it a certain way with the observer in the picture but when you don't define the observer you're just talking about a different system and so that might not be something that only electrons exhibit it might be a feature of certainty and uncertainty in measurement and that would be various kinds of measurements it wouldn't have to be just things that were smaller than a molecule okay that's how you're reading it yeah that makes sense but that's my non-technical um physics for biologists take yeah any thoughts on this blue or we can go to the keywords so i do have a little bit of thoughts and so it's tied into the quantum but it's also tied into back in theorem 7.1 so you had skipped over this word that i think is kind of critical and important um in the previous bullet point that says our main form of result theorem 7.1 shows that intrinsic can always be associated with non-commutativity so like in that non-commutative that means like you can't construct the big cocoon diagram right like when it's not when it doesn't commute so i think that it's important to um highlight that that kind of contextuality um is like you know where they were describing the context like for uh something that's beautiful like you know a beautiful what right like a beautiful leaf is not the same as a beautiful person if that leaf looked like a person it'd be a really ugly leaf or if a person would like leaf it'd be a really ugly person right like so so beautiful it's always good like that context dependent a beautiful what like beautiful does not always have the same referential um meaning and so i think that this um is is really important and ties right into the quantum aspect of things right like how they were talking about the alice and bob um with the machine right like so it's always um you know when you can measure like it's always what is explicit context right so the quantum is tied into explicit context like when we have a set of variables that we measure for every circumstance at precisely the same exact time and under precisely the same conditions that is extrinsic context right and it's different and separate from intrinsic contextuality sorry i just wanted to point that out you might be saying something that i'm yeah it's great thank you blue it's um you might be saying something that that that i'm already about to say and um but or maybe just triggered my thought but yeah this intrinsic versus extrinsic the way what it made me think of was that um you know one thing is quantifiable and the other thing isn't it's like a gradient that's essentially what you're saying yeah okay it's great thank you fun well let's get to the keywords this was definitely one of our um you know even though you look at the keywords you think oh it's the abcs i mean how hard is it gonna be no it was really fun and a good learning experience for all of us to go through these topics and includes active inference and then bayesian inference which are things that we've definitely talked about before as well as markov blankets a little bit however the other words were basically all just even new to learning about not just in active inference so that's um where we kind of started this one and that's why we want to just include people in the conversation and just see where everyone is at in learning these topics so choose space channel theory information flow local logic cone cocoon diagram frame problem and then it wasn't a key word but it's kind of related is concurrency and concurrent processes and i just always want why is the second c capitalized let's motivate the paper um not with what chew spaces are which is how the paper begins if you were to read it but let's kind of motivate it with the problems and the context that we're looking to be applying in which is the frame problem and active inference and thinking about something that most of us are more familiar with which is bayesian computation so the frame problem is the problem of using their words the problem of circumscribing what does not change when an action is performed viewed broadly it's the problem of circumscribing what is relevant in a situation it's kind of like if you want to say well what are all the connections in the social network it's the shadow of what are all the non-connections because those two answers are arrived at at the same time and if you have no false positives and no false negatives it's the same to know what has changed versus what hasn't changed so when we're thinking about scenarios those two are equivalent um instead of just reading this for a first pass um what did either of you how would you summarize the frame problem or what is something that makes it valuable or important for study okay think about it and then raise your hand while i go through these the issue with the frame problem and the reason why people are working on better solutions is that let's say we want to understand uh what does or doesn't change when action is performed so we're making a self-driving car and we want to understand how something changes or not when the scene changes um in order to look for all the changes aka what hasn't to change it's really hard to know when to stop or what has changed there's just an exponentially large number of things to check and you can't brute force it so there are heuristic solutions so solutions that aren't exact but get you on the right track however the heuristic solutions which often work extremely well like frame differencing like if something doesn't move between two slides you just cancel it out and if it does move you're going to do some other algorithm on it it's a good heuristic but it assumes coherent context it assumes that what is observed could in principle be expanded to include all that there is to be observed so if it's a image and it's a 64 by 64 image and i do frame and we're only studying the image then i have captured everything that's changed but when there are systems with intrinsic contextuality this coherence assumption is violated if there is intrinsic contextuality i.e some of the observables are non-co-deployable the current context cannot even in principle be expanded to include everything that's observable so blue what do you think about that or the frame problem so one thing that um i thought was cool about the frame problem is like that they um so they had this corollary 8.1 i don't know if you have this up there but when they're when they're discussing the frame problem it says a distributed inflammation information flow system can be informationally unencapsulated only in absence of intrinsic contextuality but so so what that really meant to me it was they said again they reiterated practically that this means that proving the absence of contextuality in a domain requires discovering all of the information that is relevant to solving problems in that group in that domain so the frame problem can only be solved if it has already been solved like it's like having ground truth data like if you're trying to predict something like you can only predict if you've been given a ground truth to to predict from or by um do you need me to repeat that danielle you caught it i just yes interesting maybe just like has it been solved like it has been solved like in the past present or the future and we're just like figuring it out again or something or what does it mean right like so it's essentially like you can't go you can't go forward solving it and this goes into and i know we're gonna get into the computing analogy later on in the in the paper the reservoir computing but it goes into that like without ground truth data your neural network is useless like like you can't predict the future based on something that if you don't have the the context for it right and not this intrinsic contextuality but you have to have like like a real contextuality like know all the variables that are important for the for the problem um anyway sorry that's that's my thoughts on the frame problem cool and at the bottom here um how would domain specific or general solutions or simply better heuristics lead to real world implications well we can probably think of a lot of software and hardware and hybrid systems that need to ask themselves what is relevant in a situation so that's going to be a recommendation engine a self-driving car whatever it is we want to be modeling relevant aspects of a sim of a situation so that is the big and it also has implications that each person is going have their own take so you know put it in live chat or the comment but it's a cool topic of course there's going to be new places of bias that'll be kind of interesting too yep any other thoughts on frame problems sarah no that's it so bayesian inference another big topic but um we can start it with a fun little meme so on the top right it says okay gang let's see what deep learning really is and this is like scooby dooby doo where are you uh meme genre and so we're gonna find out what deep learning is at the end of the episode we unmask the villain oh it's rev bass so this is sort of hinting at the idea that bayesian inference or bayesian thinking bayesian statistics is underlying what deep learning is what is bayesian inference so there's two ways we're going to show it here there's the equation in its sort of um most simple form on the top left and then a more colorful unpacked version that's made a little bit closer to natural language there's equations and then also we can look at it in a graphical way and what the equation as well as the graphical um representation are getting at is the idea that there's a prior which means before and so there's a prior expectation and it is some type of distribution it could be tight or wide it could be different types of things and there's no uninformative prior even if it's flat like i'm not sure if it could be a zero percent or 100 chance or it's all equal that's not the same thing as unbiased so this really gives a better lens to talk bias and implicit or explicit bias because we can actually talk about priors being updated rather than who is or isn't coming to the table with no prior expectations that person doesn't exist priors um are transformed through measurement into posterior estimates and so it's like reality is providing stimuli and then the prior gets dragged along either a little bit if the learning rate is low or a lot if it's going to be an overwhelmingly powerful stimuli it gets pulled a little bit or a lot to the data and um what else would you say about that blue um nothing else i mean i think that that's that's the deal um pretty well described what about you sarah i mean the only thing i thought it's a little vague um i mean i was just looking at um some youtubes on koopman embedding and model discovery and and when i see these bays diagrams i'm often like okay well they have some data where they presuppose you know what what is connected to what prior and with the uh with this this other method that i was looking at it it seems like that's not the case you know they just feed it into this gigantic structured ai um but so i'm just thinking about bayesian inference in a lot of different ways cool and we can also ground bayesian inference in a history and a developmental trajectory of statistics which is to say more memes so first here's just one quick meme here's bayes saying i thought the volume of a high dimensional hypersphere would be evenly distributed throughout just funny stuff truly funny yeah exactly and on the right side is a smbc comic just joking about how when data comes in your model is updated but that doesn't mean it's going to be an adaptive model so there's a lot of ways to go about integrating new data into priors you could have fallacious priors you could fallaciously update you could make the wrong measurement you could interpret the measurement wrong so saying we did it bayesian is like saying you know i built the car out of metal it's like what else do you want to know does it work is it safe so we can't let the buzzwords um confuse us we just want to follow up with what the buzzwords are and here in the sort of stereotypical um human evolution meme there's um with a little extra labeling there's a progression from the and theta here is the model parameters or just the model and then x is like the data or the observable so the a priori homo apriorius just is walking around with a model here's how the world is i what data just it's the model this is what's likely homo pragmaticus is sort of again it's not an actual evolutionary or creationist um parallels just to me the data is the only thing that's considered what's the data show me the facts facts don't lie facts don't care about your feelings it's just the facts it's not even about the model and then we get to homo the frequentest so the frequentist is asking the statistical question what is the likelihood of the data given the model so what for example um a p-value is coming from like a t a t test is like what is the likelihood of this height data given the model that the two classrooms don't have different heights and so you can say that's 0.01 p value so there's such and such um likelihood for this data under that model so we can reject that model and then that's all this whole point with the rejection of null models and h naught versus h1 and rejecting things and the whole idea of falsification at the statistical level but also at the philosophy of science level oh one beautiful piece of data could ruin the theory and you just can disprove it this whole idea of disproving you have the model in the background explicitly or implicitly and then new data comes in you go is this likely or not and then if it's unlikely you think that it casts doubt on your model homo sapiens here is thinking about the likelihood or modeling both the data and but then the sort of advanced or at least farthest right homo bayesiansis is thinking about the likelihood of the model given the data which actually is what we want to know we don't need to know the likelihood of all board positions and all possible chess strategies we want to know the likelihood of certain things happening given what we're observing which can often be quite limited or confusing so not to over analyze meme but yeah okay so can we wait i want to interject really quick so we were talking earlier about um frequentest and bayesian statistics prior to pushing the you know broadcast button on the live stream and you know i was thinking about the movie 50 first dates right like i don't know if you guys have seen the movie but she wakes up every day and can't remember anything except like before her car accident so the model's there it's existing right it's like she has the model but and then just just the model never updates so every day like you know it's you just go based on the model but then there's no there's no update and so like you know the bayesian would be like if every day you updated the model while you were sleeping at night and i think like you know that's a huge like like lead-in to the bayesian brain like framework right is that like what we're doing while we're sleeping really are we updating our model like is this why we need less sleep as we get older and we get stuck in a decade like you know you're stuck in your favorite decade because you're not updating your as much or as well i don't remember the movie too well but didn't the other character have memory across the dates so it's almost like an alternative title it could have been you know when a frequentist and a bayesian date 50 times that one definitely got left on the uh clipping room before so um bayesian networks are taking this bayesian inference idea and putting it in a graphical or a topological framework so it's representing variables in the nodes there's other graphical models but just for the ones we're showing here the nodes are random variables and then conditional dependencies are shown with arrows or with edges so this is like a very general one on the left it's like a is just having a conditional dependency with b and it's directed in this case this one right here is actually a bayesian network called a neural network this is a single layer neural network where one input kind of unfolds to all these hidden layers just one here and then it goes to an output layer but this is the structure of neural networks so everything that we're talking about like neural networks all kinds of deep learning are encapsulated and then on the top right from par at all 2018 is the example of active inference and so here is the progression that we walk through in our model stream but the simplest bayesian network that we can imagine between a hidden state and an observable is you have s the hidden state and then there's some sort of p which is the probability of an observation given the hidden state of oh the observable so that's like hidden state and then you're getting observables and then you're trying to model the hidden state given the observables in b that same process happens through time a changing hidden state is emitting observables at different time steps still a bayesian graphical c introduces control theory with pi that's why our lab logo is a little pie because it's action playing into the way that states are modeled to evolve through time d takes another level of complexity and allows actually policy so still we see this motif with pi influencing how s one two and three change but at each time point let's take this one down into s1 at each time point it's being evaluated how a policy is going to play out at s1 2 and 3. and notice that at one two and three there's an inference on one two and three and so um i'll re-enable my camera don't worry the uh the trajectory of action is what is being estimated so it's not just doing state estimation at each of those time points it's estimating a trajectory of action and this idea of action trajectories is going to come back into play in a kind of interesting way let's just take one more second more generally to think about active inference and what it is why we're here where the markov blanket is in active inference we have internal and external states and it's a scale-free framework because these internal states could be small or large in scale it's outside the level of specifying a specific scale and these internal and external states are separated by a blanket and as we're going to show in just a second this markov blanket concept is just the total layer of insulating nodes such that the internal and the external states have a certain type of independence relationship from each other and then the innovation we'll get to where first thing came into play so that's active inference just um as a way to integrate action and perception um with sensations influencing internal states internal states including generative model estimates of the world and policy estimations implements action and action can influence external states which also is undergoing its own dynamics which we might think of as actually contextuality any other thoughts on active inference okay so here's where we go from bayes networks to markov blankets this was part of our discussion on actin 14 and the idea of making a markov blanket was introduced by pearl 1988 and where fristen's innovation occurred and this is i think from mel andrews's paper in 14 is that pearl's original construction of the markov blanket was subdivided into two kinds of nodes sensory states nodes whose influence is directed towards the blanketed node under interest x and active states which are nodes influenced by x so we can have an organismal metaphor with sense and action but a little bit more generally sense is like incoming of the system and action is outgoing and the markov blanket is the total set of all the ones that totally give you all the inputs and all the outputs of the system so stating the conditional dependencies is the same thing as stating all the conditional independencies just like we're showing where the social network who's friends with who is the same question as who's not friends with who who is there is the same question as who is not there and then to put markov pearl and fristen on a continuum you have markovian properties by andre markov and other mark hopes actually who are working in a more mathematical frame they don't have access to computers perl brings in bayesian statistics as well as computational approaches in the second half of the 1900s and then the innovations of fristen were to divide those markov nodes the markup blanket nodes into sense and action nodes which enables this cybernetic approach to input a generative model internal states and then outgoing uh action states in service of maintaining a non-equilibrium steady state and then though it's an ongoing and open and developing area with how we formalize these things the idea is that by this kind of partition of the blanket happening we'll be able to have um modeling of non-local dependencies like latent causes which are not observed in the world and therefore have cybernetic agents in niches that are enacting and embodied and encultured in a really rich way and then just to show one more figure this is from um wanja's paper with carl fristen which we talked about just a few weeks ago and just showing that you have internal states that influence external uh influence active states then there's a line with external and so here's the blanket here in the pink dashed line is the blanket states and then there's the internal states okay any thoughts on active inference or can we go to context so again putting the sort of end parts of the paper up closer to the beginning the context in practice the first context introduced is the idea of active inference in the paper and the motivation for active inference here and we can definitely talk to the authors more is that observers are constantly probing their environments through action influenced by hidden states and time-sensitive policies so these are some of the challenges that beset context agnostic or unspecified context modeling is when the environment is changing how does the action policy update does it just re-evaluate at every single moment as if it was like fresh looking at a new environment but how do you have these sequences of action that might involve sequential modifications of the environment and uncertainty in the action policy and the big question just like you know kindergarten is what actions are appropriate what actions will reveal aspects of the environment important for defining and reasoning within a context that's the frame problem what actions will reveal that a context has changed so we want to know which of the three cups the ball is under and so there's like this complex thing that's happening as we guide our action by changing the context to resolve our uncertainty about actions which we can then get more information on and active inference tells us that there's two ways to minimize expected free energy which is that we can modify internal state parameters of our generative model through learning and development or external states can be modified through action or um their own endogenous changes so active inference we're kind of going even more abstract than some of the other papers and just thinking about this question of if you're an agent who's operating under uncertainty modifying the niche but you don't know whether you should modify or explore or wait and all these things are coming together we want to set a frame that's big enough to encompass that any thoughts on that okay let's turn to context more generally so that was active inference in context they write contextuality in the behavior of complex systems has traditionally been regarded as a practical problem of limited experimental control that could in principle be eliminated by obtaining more uniform experimental subjects and achieving better control of experimental conditions this traditional view of contextuality as merely a practical limitation is howeverly increasingly under strain this kind of resonated with me as an experimentalist because in biology you'd get the sense like well if it was a clonal animal and we just had the light dark cycle the same and the same animal handler picked it up and there was no pipetting air everything was the same you'd get the same result and it was really exciting to think um to what extent is that like a faith-based position and actually tiny amplifications in the system you have two clonal animals but if there's like a bifurcation point then unless you truly have every single part of the system you're always going to be subject to these intrinsic contextualities so that's kind of cool to think about and then here in the conclusion context is often used to designate what is neglected when an observation is made or an action undertaken environment is used in a similar way so that's how maybe informally people think about context like right now one of my contacts is like this cup of water it's in my environment it's in my niche it's in my local area like their context is used like that like what is around something um and the formalism of a theory like context by default challenges us to acknowledge that context is always there whether attended to or not so we're thinking about context in sort of a more active way in terms of what influences processes or where conditional dependencies are and so it might be the case that you can ignore context because you've actually isolated the system so it's like a closed system or it's a system where you can fully describe it and in those situations context is there but you've made a special sub problem where you have all the context but when you're in a situation where you don't have all the context we need to have a framework for reasoning that has contextuality by default so that we can work better in those spaces and let's talk a little bit more about this contextuality by default cbd not that cbd this is the authors talking about contextuality by default just totally you know raise your hand or jump in if you have a thought or a question and they say that the term um determining whether a particular instance of observed behavior of a complex system particularly one with memory exhibits intrinsic contextuality is not straightforward so that was interesting makes sense complex systems hard to know what to do sometimes whether you want to include memory and historiosity or not and in some ways like if you have like 10 different animals and you think okay well they're the same background so we can just consider them even but they've actually had this history that differs so they're not the same but you want to imagine that they're replaceable or shuffleable in a statistical sense but that might not be and then here's the quantum sense of in quantum systems a set of measurements exhibits contextuality if it cannot be characterized by mathematically consistent context-free joint probability distribution asking whether a set of measurement outcomes exhibits intrinsic is in this case asking whether a consistent globally defined globally connectable joint probability distribution exists or not and that's where this formulism of addresses questions by labeling context and includes them explicitly in conditional probabilities rendering probability distributions so let's look at the papers that were cited and referenced and think a little bit about where is this contextuality theory coming from where is it taking us what do you think oh yeah comment um that that brought to my mind um emotion uh related to context because i've been thinking a lot about um how like if i'm attached to some idea or something like that and then and i just remember that i was attached to it but then i go to sleep and whatever and then i forget what it was i have a tendency to like really be like no that was really important and so regardless of whether it was true or not i want to go back to it and so there's i'm thinking about the way in which emotion is a kind of a context but it takes you over a longer time domain because like it doesn't even just sit in like a decay rate that would normally be you know related to the other information it pulls you way back into another space so it's this kind of like long-range driver i don't know that's just context cool and it brings us to this next question which is is there this is the title of a paper that's cited by um this offer of 2016 is there contextuality in behavioral and social systems so if we were dealing with this informal definition of context is just like the environment it's like yes the answer is just yes behavioral and social systems yes there's social context like oh you missed the context on this dinner party okay so off the bat we'll say yes but we're thinking about it formally and this is this is the most hyphenated sentence i've just ever seen in science so just the number of names it was quite something to read um but we're thinking about contextuality in a more formal way which is like so let's say someone says you're missing the dinner party context um and then it's like well if you would have told me all the info then i would have had all the context so then that would make it that's it's a similar notion but we're going for measurement systems and um they're proposing that their cbd allows one to define and measure contextuality in all systems of all these hyphenated types even if there are context dependent errors in measurement or if something in the context directly interacts with a measurement these are the two cases that really matter a lot for scientists like what if at higher temperatures your thermometer is biased towards over estimating or being more error prone then you're going to get this distribution that looks like it has some sort of super specific relationship but it's totally not the case it's all an observation error and the second one is that the context could directly interact with the measurement aka every single time you want to know something new so the context which you don't have because you don't have the full context because you're going out into the unknown to make a measurement is going to interacting interact with your measurement maybe even modify it itself what do you think blue so i think like you know this just goes back to that intrinsic versus extrinsic context like in a social and behavioral system there's no extrinsic concept right like every er there's no extrinsic context every like thing that you're going to try to measure if you ask like what was the temperature at that dinner party or how is the food you can ask like you know 12 different people and they're all gonna have different uh things to say or like did you have fun or or whatever like i mean everyone you ask because it's it's only defined by the intrinsic which is the experience of each agent in the room i think cool and interestingly they look at several data sets none of the data provide evidence for contextuality so again don't just yes there's a niche there's a context we know what social context is but we're actually thinking about it in this very interesting way where those data sets don't have contextuality so a good little gut check is if you're thinking about the applications this paper and you're thinking about it like social context it's just not that one because um it's not and their rule is that behavioral and social systems are non-contextual again if that makes you cringe it's because you're thinking about it informally and we're actually thinking about it in a way where that claim makes sense that's where we want to understand to be like oh right informally we get where they're saying it but then we're interpreting their claims rigorously i.e all contextual effects in these contexts or in these settings result from the ubiquitous dependence of response distributions on elements of context other than the ones to which the response is presumably or normatively directed so let's say you have a driving simulation and then there's going to be some green lights and some big red stop signs and you do the behavioral measurements and you find out that conditioned on whether it was a green light or a stop sign the people accelerate or break so that's your experimental data what you controlled and what you observed was stoplight and then whether they stayed still or accelerated now that is context um it's it is context for the person who's driving that's the social sense but all the contextual effects in this experiment result from the ubiquitous dependence so across the whole population of response distributions on elements of other than the ones to which the response is presumably or normatively directed so it's like you thought you were measuring the response to a green light or a stop sign actually you're measuring a social and a group of people who were in a relation with those symbols such that they operated in a certain way so we need contextuality by default so that we don't fall into like correlation causation super fallacious scenarios here's another case of where context matters and what's going to be fun is actually this decision making is going to matter for context and arrogancity here's a paper by valve um local choices rationality and the contextuality of decision making so every decision has a context so we know about that however rational explanation appears to be challenged by apparently systematic irrationality observed in psychological experiments especially in the field of judgment and decision making so how many times have we heard oh people are so irrational they'll take the one dollar today rather than the dollar 20 cents tomorrow or they'll pick up their children from the daycare at the right time unless there's a monetary penalty and then that will make them later they're so irrational within this one variable that's being estimated here it's proposed that the experimental results require not that rational explanation should be rejected but that rational explanation is local i.e within a context thus rational models need to be supplemented with a theory of contextual shifts makes a lot of sense um it sounds like something that's relevant to learn a bit and this paper has probably other relevant things but it's like instead of just saying well who's rational and who's irrational couldn't we just understand that everyone is in a local logic and a local context and then they're operating a certain way conditioned on that and so it's not a rational versus irrational divide it's what's the context who's the agent what's the niche what's the context shift so that's a much more open but also powerful way i think to talk about okay just raise your hand if you want to pause it let's take this idea of context so we started with context in the sort of um folk sense and took it to this idea of defining all the relevant variables now let's think about defining relevant variables in a model in terms of topology so here's some cool slides from the simmons institute and in this top right one it says topology is about distinguishing the continuous from the non-continuous and also about moving around so i was that's a cool definition of topology not nodes and edges not how things are connected but it's a way it's saying how things are connected and this slide was really informative when we're asking several questions at once the answers could obey constraints so like if you're going to measure the pressure and the volume and the temperature of an ideal gas there's an equation pv equals nrt that's going to relate those things and if one of them is out of whack like probably something is wrong so those bundles of answers are related to each other could be by the laws of physics like here's uh you know velocity and time or whatever it happens to be like physical constraints or logical constraints so like if things are true then if they're false if they're not that and then if it's double not that it's back to being true so you can have these bundles that are logical or a bundle that is based upon some of the constraints of physics and then also the constraints can be rows of a table in a relational database that's the trip because that's where like sql and a lot of programming comes into play and models distinguish good and bad ways of connecting the dots and bundles just like continuous sections so this is kind of like thinking about sql queries and database queries in terms of the context um that's being concerned saying in the context of this database i want to make a query within it so that's an interesting way to think about how things are connected and context but um this is also stuff that's we're all just trying to put out there and hear people's reflections and thoughts on what do you think sarah i just wanted to restate it to make sure i understand it um i mean basically say they're saying the the the the way you move through context sets creates the topology space i think that's is that an accurate way to phrase it um just another aspect of it yes like there's certain in the space of making multiple measurements those measurements can have different topologies like if you make three measurements they could be in a line they could be in a triangle like there's that's not at the formality that's on here but when you're making different they can either be in a well-behaved bundle or a not well-behaved bundle and then watch whether it's well-behaved or not is um related to whether it blew what do you think oh yeah i mean it's a hard thing let me just i'll jump in for you so so um going back to like the intrinsic and extrinsic um contextuality like so it's contextu it's contextuality when you have like a set of measurements that you know um are corresponding measurements right like so i mean it's like um i don't know do you have bio background sarah like like i'll i'll talk about it just in terms of like you know rna sequencing right like you can sequence your sample in a different lane on the same flow cell or in two different flow cells and there's variability right so you always have to try to like design controls into your experiment from flow cell to flow cell from lane to lane and so that's why you replicate the sample sequencing just to kind of control that that variation in just running it at a different time the temperature in the room might be different or you know the machine might not be feeling as well that day or or whatever so so it's these kinds of are the measurements are they all like do they all link up like is there like a correlation between all of the measurements or not so that's the well-behaved bundle is when they do all link up they have that shared um information right so also i think as we get into the paper maybe it might start to make more sense yeah okay let's go to channel theory okay so this part was super fascinating um and i'm drawing a little bit on one of the background papers mosaic of two spaces and channel theory one by fields and glazebrook which we're going to talk more about the two mosaic papers soon but let's contextualize this in terms of information theory so-called info theory but i personally call it dis info theory and i've called that for a long time and now i know why actually so here's the second paragraph of claude shannon's 1948 work shannon writes the fundamental problem of communication is that of reproducing at one point either exactly or approximately a message selected at another point wait is that the problem of communication am i trying to reconstruct something bit for bit via communication or is the goal of communication to impel joint action to update other agents not to replicate a message shannon's working with telegraphs and copper wire so it makes sense why you want high fidelity data transmission a signal of transportation of information but let's think about how broad people have gone with info theory according to shannon um and then he writes frequently the messages have meanings they refer to or are correlated according to some system with certain physical or conceptual entities these semantic aspects of communications are irrelevant to the engineering problem o semantics are relevant to communicating information semantically for people or for what the significant aspect is that the actual message is one selected from a set of all possible messages the system must be designed to operate for each possible selection not just the one which will actually be chosen since this is unknown at the time of design so fair enough a through z you want to have all of those options because you don't know what next letter needs to be poked but i think if people read through shannon's work and considered a few of the aspects of the shannon entropy like how it disorders messages like it just asks what's the symbol frequency of the whole message or page or genome if you think about how it removes the ordering of symbols and is actually asking about replicating bit for bit messages not communicating in any sense people would really consider what information theory is thankfully in this paper the fields of glazebrook wrote about comparing and this was cited in the 2020 paper comparing and combining the shannon theory of information with okay so um though very general as a quantitative theory of communication the original shannon theory had largely overlooked the question of semantic content in any dretski type theory the basis of semantic content is in the world i.e the events or situations that signals or states carry information about the car you know it's out there it's in the world by showing how local logics are connected by information networks channel theory provides a generative general qualitative theory of information flow in this context so that's pretty interesting it's more information flow that semantic rather than syntactic and then this last part here um this is there's more to say here but all while in 2004 create a synthesis of shannon's quantitative information with the bar wise seligman qualitative to address how the question of how specific objects situations and events carry information about each other that's what people are always talking about when they want to be talking about sharing information not sharing surprising symbols but updating each other semantically informationally in the terms of bayesian priors about the world things that we're referencing in the world the temperature not just as a variable but like as about something in the world so some thought questions is shannon 1948 correct about the stance taken on the fundamental issues of communication or what communication is or isn't how is channel theory distinct from information theory so called this info theory we can start calling it and then how is real communication the kind that we care about and want to model in many cases how is it both syntactic and semantic all these things at once grammatical logical as well as lexical word-based it's also narrative driven a narrative or the narrative context can make just one word semantically powerful speak now or forever hold your peace it's a narrative where one word cannot be reduced to just the bits or something like that oh well but no is a common word or yes it's a common word it doesn't matter that it's a common token because it's semantic information how is it contextual blue so i mean i think that there's this of i i like shannon information theory i like how it's very logical and it appeals to like my my my brain structure um the semantic information that's contained in this channel theory description is very squishy uh and and i think maybe um not as appealing to me but i do think that it's more correct right like when you say when you reproduce the information like if i say um something like oh it's raining outside i live in the desert so like it's raining outside it's like a really awesome like event like let's go play in the rain right so when i say that and someone else says oh oh it's raining outside like like oh i'm like my plans are you know destroyed because i wanted to go to the park and i can't because it's raining outside so so it's two it's the same information but like i the way that i said it was very different in those different contexts right so cool let's return the discussion from information flows and communicating information and we're kind of thinking shannon syntax information reproducing bits bits on this computer to bits here we're going to take it up a level to semantic information and then semantic information so like i'm thinking of a purple cat now you're thinking of a purple cat but that could have been in a different language and it's not really related to how surprising each word is let's return to this question of local logic and contextuality so here's um a paper not sure if it was cited or not but it's going to bring together local logics contextuality and topological approach so we're just going to kind of give multiple codes of paint on everything local logic is like the local rules or the context and in this um kishida 2016 paper they wrote the topological approach characterizes contextuality as global inconsistency coupled with local consistency so that was kind of cool to me it's like if you have this field and it's globally consistent then it's like transparent there's there's nothing inconsistent to speak of there's no fracture in the glass so it just is purely transparent so um that's just the same it's part of the same context but local consistency coupled to global inconsistency is like a bubble it's like it's an inconsistency in the tank of water but then locally it has another transparency that makes sense in that local context and so you could have like a computer program and then that could be a local logic but it's embedded in another logic of being a turing complete or you know whatever kind of computer and then within that program there could be like a simulation of another program of an n64 running another program and even if it's all the way back up at the top it's running on linux but that local program is running on the n64s local logic and then another kind of metaphor oh and then in the paper they write our goal is to capture this local consistency part not just the global inconsistencies the global inconsistency is like our receptor is not getting any photons boom it got a photon inconsistency so not just that part but the local consistency part which so how do we know how do we know that it was consistent before that photon hit which requires a novel approach to logic sensitive to the topology of contexts to achieve this we formulate a logic of local inference by using context-sensitive theories and models in regular categories and that's going to be powerful here's how i was thinking about this in terms of chess so the rules of the game of chess are which piece can move which way and then some of the imperatives like you have to um alleviate check otherwise you lose the game that's the context the local rules of the game then there are local or situational patterns like a pin where one piece is like in front of the king and it can't move out of the way because then the king would go into so that's not a rule of chess but it's an emergent pattern that arises from constraints in the space given the local logic of chess but now we're in this even sub-local logic i'm not going hyper formal maybe chess players have another way of doing this but it's like there's a local logic in the corner of the board with a king and not every piece or rule of chess like okay well the pawn can move two or one but that rule might not matter for this pin with these pieces and so not every single rule or piece is going to matter for these local logics and maybe there's a way to pursue this topologically in terms of the connectedness of different relationships of the system let's take that to choose spaces okay so this one was um you know it's the second section of the paper so it's like okay you gotta understand and and learn this one so we were kind of excited to do this so they start by defining one of the simplest categories so we're in category is the category of two spaces which we're going to talk more about and it basically is about a set k and then within k there are kind of two kinds of things objects and attributes and the two space is like a matrix that is the relationship over that set k of the relationships of a and x it's so it's actually probably a data form that we've seen a ton of times here's ten people and then have they rsvp'd for my event or not yes yes yes no one one one zero that's a two space for rsvps but it's like that you know here but it could be a lot more general and we're going to keep on generalizing on that but the basic two space is a matrix but it probably can be higher dimensional that is just a regular matrix but a topological space is also a matrix so you can have like the adjacency matrix right like all the nodes by all the nodes one or zero connected or not or a weighted graph how strong is the connection or it could be symmetric and undirected or there could be a from and a two so like all of topology is matrix forms and it turns out that they're two spaces but also we can think about like programming languages and we can have a type as one of like an attribute or a type you know is the wood tabling shoe space or is the object a table that's a chew space so there's a lot of programming relationships that the chew space covers probability spaces um coin flip you know content you know that is it gonna happen or not so this it's like if it's a matrix i don't know if it's you can go as far as say it is a two space but it certainly a broad family of objects or categories i don't even know what to say there are within this two-space paradigm there's a few more definitions because we looked up just a bunch here's from wikipedia and basically the big idea is that the true morphisms are transformations on two spaces and if things are set up properly the morphism's continuous and the pair of functions has some special relationships so um one example this makes me think of is like the typographical number theory in girdle escherbach where it's like one plus one equals two and then you make a set of text rules like you can add an s to the string under these contexts or you can add um or you can you know it's like it's basically doing addition and subtraction by typographical like letter based rules so if you can prove something in one because you've super strongly shown that there's a morphism you can show that that same relationship should exist in another system it also turns out that's actually at the heart of like the whole gerdell question but that's the static interpretation and then um here's dynamically two spaces transform in the manner of topological spaces um yeah don't go too deep down here but there's special relationships of two two spaces that are chewmorphisms and then if that morphism is bi-directional it just allows a really nice thing where you can sort of make a finding on one side and then arbitrage to some relationship that might have been difficult to show on the other side so it's like a perfect language mapping you could say you know the the boy is younger than the car if you had a perfect language mapping semantically you could say it in another language and the semantic information would be perfectly preserved so then maybe one person has a piece of semantic information in english and then they could convey it through the perfect morphism and it would be correctly semantically understood on the other side okay here it's just a couple these we just we wanted to actually understand it and also um shout out to master students everywhere so here's history from this paper the chew construction takes a symmetric monoidal closed category v and with pullbacks i got reading it and an object k of e and completes v to a self dual category chu v k the details of this construction appear in ph choose master's thesis published as an appendix to his advisors mbar's book introducing the notion of a asterisk autonomous category so chapter that was an appendix and now here we are many years later so it's all good to research whatever you're researching on if it appears to you to be the big question here's barr's so bar is another perspective on shoe spaces barr had this idea during his sabbatical to study duality in categories in some depth just what most of us take our sabbatical wondering he wasn't wondering about the simple dualities but self-dual categories like complete semi-lattices or finite abelian groups he was interested in the possibility of having a category that was not only self-dual but one that had an internal home and for which the duality was implemented as the internal home into a dualizing object it's technical so duality is going to mean something different to a mathematician than a non-mathematician i don't know what a home is however searching for an internal home i just thought okay the dual it's like forwards and backwards or it's like it's it goes both ways that would be nice you know reversible it's like those are always good things and then the internal home internal homes when chained together form a language called the internal language of the category and i thought okay so now we're going from categories and attributes to like something else the most famous of these internal homes are simply typed lambda calculus which is the internal language of cartesian closed categories and the linear type system the internal language of closed symmetric monoidal categories lambda calculus is something that's quite broad and central to computation so this is like something about how logic local logics and nested logics are related to categories in an analytical way such that the kinds of logic that we see on computers is only a type of it because in a way we're dealing with syntaxed based logic you know you have two files are they the same it lines up the bits and if they're the same it's yes no but what would this mean if it could be semantically the same and how would we make that non-squishy because to ask whether semantics were the same you need all the context so that's kind of this fun twist and it why for simplicity you use frequencies statistics instead of bayesian inference because it is hard to do sometimes or you do have to grapple with unknown unknowns and it's easier to just do a t-test or a multi-linear regression rather than some other type of generative model because that takes like bravery to specify your uncertainty and to give support to why you did a certain thing rather than just say well you know the countries differ with a p-value of 0.01 but it's easy just to say that and move on rather than actually do some sort of rich modeling underneath okay true spaces and channel theories so like we're kind of just trying to bring all these two you know connect two of the ideas and then you can start nucleating some ideas from the ones that are making sense in channel theory which again we're thinking of like the um more general cousin of shannon dis info theory in two transforms these are the transformations between the two sets um or spaces become infomorphisms which are natural maps between classifiers okay so blue what would you say about the red underlined part or what you wrote at the bottom um so this will i'll like transition you into the next slide maybe here in a minute but um just the red underlines part it's the transform from a to b so if you have one two space and another two space the whatever it's like you know you can think about like a territory and the map of that territory like whatever transform going from the territory to the map that is that channel right is thought informally of as a channel um that is uh like the channel that we're talking about here in in terms of channel theory and so these um the the classifiers here so the um a classifier is the um set of where did i have that here somewhere classifiers link tokens to types oh yeah but so oh that's that's this is the definition of a classifier so classifiers link tokens to types that encompass them oh yeah because you've already read this so the channel theory the two transforms become infomorphisms which map between classifiers so if you have different sets of classifiers right so classifiers link tokens to types that encompass them so for the type stoplight the set of tokens could be like you could have a red stop light a yellow stop light or a green stop light and that's also known as a local logic the classifier which we talked about local logics just a few um slides back but flip to the next slide and i'll continue this so mapping between classifiers so here if you have the classifier stoplight and then you have like what the motorist should do with the stoplight so that's the second classifier you could have the tokens stop at the red light go forward at the green light or proceed slowly at the yellow light or like get out of the intersection or whatever it is you think you should actually do at the at the um stop light but then you can have other classifiers too so it doesn't just have to be the stop light and the what the motorist should do it's also like what pedestrians should do and like at the game like red light green light that you play when you're a kid like you should run forward run slowly or walk or or then and stop so i mean there's different um classifiers that you can have that all mapped to the same central information core right and so that's what you see here the classifiers are these things a1 a2 and you can have many of them up to a to the k and then you have at a core this channel right so this is the infomorphism that allows semantic information not just shannon information to be transmitted between classifiers so this comes down to having a shared memory so we all learn from the time that we're six that red means to stop green means to go yellow means to slow down so so but to a six-year-old it means running you know when you're a pedestrian you're looking out for other cars and and there are other behaviors that you do and as a motorist also and as a police officer policing the motorists there's all kinds of people that have different reactions to what the stoplight is doing but that's all based on this shared memory like we've everybody has been taught like this is what you're supposed to do out of light um and so so that's this core information channel um through which flows it's can be also be thought of as a shared memory or um a shared context if you will cool one thing that that makes me think of here is like this sentence that said um uh the existence of a true transform a to b is equivalent for every flow formula valid in a being valid in b so it's kind of like you have the chessboard and the board position says it's game over and then you have the the data file and then those two are within a bigger c that is establishing a connection between these very different types and so it's like there's um in the example you gave the local logic is like there is a total logic that it does all come back to even though there's disparate types and there's sub logics let's hear what everyone else and chris can bring to this discussion because it's like super interesting to think um yep okay so what about sequence blue um so the sequence here uh i just wanted to put this in because i thought it was it was important for information flow so sequence uh when a secret incl encodes a semantic like a causal constraint uh that effectively functions as a logic gate so this really what it's what enables semantic um information as opposed to shannon information so it's like you have the message but there's a logic gate there that says if the message is applicable or not applicable or to be transmitted um and also that it um i think puts forward into the um it enables putting forward into the scale-free or scalable uh infrastructure here i think um and then i like all this like math symbols like i've never seen this stuff before these like what is this like parallel line lollipop stick i don't know um but but i just i i put a box around this because it's like oh that goes back to bay's probability all right i know this so it's coming home again cool i think this one just reading this one um x then shape n to assign a probability x satisfies n so maybe it's the sign of satisfaction i guess it just says something that we don't know what it looks like um but it's like there's a satisfaction x to assign a probability that x satisfies n given that it satisfies m so it's like what's the probability that we're in checkmate on the board given that the file says checkmate well if you have the morphism it's perfect but if you don't have that then it's probabilistic or it's different so it's a conditional probability that's conditioned upon the way that you can map across different kinds of things and it was the perfect connection to beijing inference because at the bottom here we have um basically m satisfying um a is the true space that's the the scarlet letter the fancy a is the two space um and so this is almost like saying m is satisfying some true flow relationship with n that is itself about how tightly the spaces are connected and so then we can take this p m given n single line not satisfied um and it is belief updating and so then uh yeah it provides the necessary semantic consistency for such updating infomorphisms thus capture a significant representation of bayesian inference with the necessary coherence since the target classifier context emits the same semantics as that as the source within the information flow information or infomorphisms are information flows that capture the updating of something semantic because it's semantic in that dretzki meaning is in the world way like the prior and the posterior are about the world and then as new data or events come in which i didn't think we captured as much here but like this event theory connection it's like bayesian um statistics is kind of like the steady state and then new data comes in it's like an event in a network and the in that event propagates through a network of semantic information flow and then that updates the system and maybe it's like the system's already vibrating and so the new event comes in nothing changes with a vibration that might be a situation like the system semantically is already at its non-equilibrium steady state and so it's like not surprising but then there's other kinds of events that enter a system that might be more surprising but it's all about the semantic nature of a system okay so wait wait yep can i just add on to this yep but like in our in our um contextualizing like and i don't know if you were going there or not but i just want to explicitly state that like bayesian inference is the event like which we are updating with active category theory and two spaces were you like like implying that or i just wanted to state that explicitly because in this context contextuality talk about like we are updating our own like events with our with our own knowledge yep and i i think what's sort of fun about this semantic info flow idea the whole channel theory thing which i really didn't get exposed to in a lot of contexts was that instead of the propagation of bits and bytes on wires and the signal fidelity and then it's like well then where does the semantics get unpacked does it get unpacked at every little post office and they just say no no just syntax all the way down so it's inherently reductionist because you go no we've explained away meaning because we're studying the connection from a entropy perspective and then it's sort of a strong internally logical argument which is why there's people who believe it and then it's like could we actually think of semantics as the heart of the because when we're talking about communication of action or modification of the niche or solving the frame problem we don't need the syntactic differences we want the semantic differences like what has changed my car is broken i don't need to know the syntax but from a action-oriented semantic view that's what we're communicating on and then we're seeing these specialized message passing systems like on a silicone chip it's a special case it's a controlled where we've defined the context so that it isomorphically maps onto bits and voltages okay here's the two fields and glaze brook papers from uh 2018 uh mosaic of two spaces in channel theory one and two so one category theory concepts and tools is like a lot of formalisms diagrams but two has some nice images and that's applications to object identification and neurological complexity is parts and whole relationships and this is to go from the two spaces to this cocoon a system capable of both object history construction and its dual category learning so just there history reconstruction and its dual category learning it's like i'm learning about cars being broken or and then if you tell me it's a broken car i can generate a vision in my mind of a broken car so it's like you have a bi-directional relationship that's that's open to generating specifics from the abstract or updating the abstract from the specifics which is really related to this hierarchical bayesian learning and active inference and they have reached after many years of research and understanding that objects that have this sort of analytical relationship between object history construction and category learning with a singular category of maintenance as a special case not sure i think that maybe that has to be one object that does both a self dual not just two objects one that does one part one that is the other is characterized by a cone cocom diagram a cone cocon diagram captures the simultaneous upward and downward flow of constraints that characterize human visual object identification and it is reasonable to suppose other sense modalities and they then take it very instantly something fascinating which is the functional duality between a high precision expectations and high precision inputs in an active inference and hence the duality between dorsal active and ventral passive attention so it is about this precision modulation and the way that you get high precision expectations and high precision inputs so how do you have a clear generative model of the visual field but then at the same time you want to be able to recognize new objects at high fidelity as they come into the screen come into your field so how are you going to have high precision generating and high precision learning and there's going to be this rich um ongoing integration of priors and new events and posteriors that's semantic so it's like i'm looking at oh it's a face and then my eyes engage in an action selection pattern appropriate for a face that reveals some sort of anomalous information pattern and all of a sudden the meaning has changed and so it's almost like what is crossing the interface perhaps is the the data the observable the but what is crossing is actually like the reconstruction of semantics rather like if it's a car driving across your view but it's going 5000 miles an hour and you can't see it then for you as the measure it doesn't matter because you didn't see it so even though it was perhaps there but it's just a blur and so this is just getting at something very interesting which is these systems that are exquisitely generative yet also hyper able to recognize and to update the generator based upon the specifics of which have been generated oh all cats have stripes all right you just saw one that had polka dots and now you can generate that and then maybe you might generalize and even generate another type of pattern of cats and then you could see whether that was accurate or not so here is the diagram that blue talked us through here with the core information channel c at the top and so this is the cone diagram on the top it's it's like a pyramid diagram but maybe if it rotates it's like a cone or there's some other way of thinking of it like a cone and you have the classifiers and this is the whole choose space as classifiers scarlet letter a choosebase so the classifiers are linked through infomorphisms semantic bridges to the core information channel so like here's the chess board here's um another two space and here's another two space for chess and they're all linking to this like game of chess local logic and then there's a map between the board and the computer program so g one two two three are the commutivity so this is like you got a link between the board and the computer and the computer and the drawing and they're all infomorphisms to see the game of chess so semantically if there's a checkmate on one board it's like a checkmate across paradigms now when you're um when this is a a tight cone and it represents like a huge variety of things and then i'll get you the second blue the co cone is when we oh yeah go ahead go ahead good let me let me interject before you do the cocoon that's why i was like okay so i want to interject here for sarah cause remember we were talking about like the the observables being like a tight like a tight little cluster of of measurements versus a not tight cluster of measurements and so here they define it in this before we go into the cocoon um so so you can kind of take this back to like the stop light and the kids playing and what the driver does what pedestrian does and whatever but but here if you think about it instead of like that the categories or the sorry the classifiers uh it says when we define a finite information channel c uh as a finite indexed family of infomorphisms those are all the f's having a common core the component classifiers will be taken to a system of observables or some subcomponents thereof so here they're specifically stating that all of the classifiers here are all of the things that can be measured or observed in a in a context that's extrinsic as opposed to intrinsic i think so i could be wrong i hope that someone someone corrects me but i feel like this is where where it's going cool cone diagram then we introduce the co cone the shadow as above so below wouldn't have it any other way a commuting finite cone cocoon diagram which one's the cone which one's the cochrone right comprises both a cone and a cocoon on a finite set of classifiers so they're mapping to the same set of a classifiers the same two spaces are being bridged and um then you have like sort of two looking at the same bridge and then uh if there's a cumulativity across the elements of the of the two we have that from above and then there's infomorphisms from all the a's to c prime and all the a's um to d prime then it's a special kind of object what is special about it it captures in addition to a ton of other stuff probably a more subtle duality remember the whole thing about studying objects that were dual objects from like bars 1975 paper and chris fields other research like what is a dual object shadow a more subtle duality between processes so not an object duality but a process duality maybe forward and backwards it enables object files object tokens and object histories to be viewed not as tokens but as types that organize respectively trajectory components that's files features that's tokens and feature based singular histories into mutually consistent collections so this is some next level token algebra and i wondered if it related to crypto because it's like instead of the history like the object history my web history well that's actually a feat that's a singular category it's a singular trajectory if you take this list of a million things and it's like one of a unique kind of thing and then even objects and tokens and descriptors are all understand under being understood as highly relational highly semantically mappable way so that's pretty cool and i guess we'll look forward to seeing what the authors can help us with or like what system could we apply this to first or what would be an empirical system where this is being measured or what would we gain by measuring something this way yeah what are various examples of context um and then here's just a few figures from that other paper the number mosaic two so here's a cat well it's an image it's a pattern on a google slide who's to say it's a cat right that's semantic but there's all kinds of things that we might want to have so it says an object token is classified at construction into multiple types by distinct but cross-modulating processes this includes animal animacy and agency detection emotion-mediated threat detection and entry level followed by super and subordinate categorization into types of objects so it's like depending on the context and the agent observing this stimuli whether it's a predator or prey whether it's a cat or whether it's felis domesticus or whether it's a sub breed of cat whether it's something that you approach or something that you withdraw from or something you associate culturally with good or bad luck that is like the agent's ontology as they interface with a cue in the and then this is a funny image it's mapping chris fields brain so it's like it's it's it says like cf that's that's him so here's cf is a human humans has a brain and then here's like another ontology of brains a private brain is a mammal brain is a vertebrate brain is a brain so you can go superordinate and subordinate categories in a relational ontology as well as have these like totally disparate classification schemes just like you could in programming so you could say this person has a height and a weight and that's in their public health record and then there's this other thing are they a friend or foe and that's my private information and then all of those are being linked in a special kind of system all right that kind of that's one little section here is bringing in this element of concurrency so some people will have heard about this from programming languages like golang or from other kinds of concurrent programming but this is pretty interesting because it's from 2005 and it's called paper by pratt it's i think cited in glazebrook and fields two spaces and their interpretation as concurrent objects so it's yet another way that we can think about two spaces not just as mapping attributes to types or doing all these other transformations but like also as concurrent programs so they write here that the two pressing questions for computer science in 2005 maybe forever are what is concurrency and what is an object the first question is of interest to today's sibling growth industries of parallel computing and networks both of which stretch our extent models of computation all well beyond their sequential origins so you have the turing tape and it's just zooming backwards and forwards on a tape it's a standalone machine it's not connected to the internet when you have machines that are in networks then there's semantic logics that are happening due to those local logics that are boolean but at a higher level they might be more like a complex adaptive system like a ddos attack on a network or a certain type of governance game uh situation in a crypto economic network using a cad cad model those kinds of complex adaptive dynamics at the network level may or may not actually um be able to recapitulate from the syntactic or from the lower level and so that's the first question is how do we think about um about concurrency and about designing for good concurrent systems when our parallel and distributed computational and even unconventional computational paradigms are rapidly expanding the second question is relevant to programming languages where the definition of object seems to be based more on whatever software engineering methodologies happen to be in vogue than on the intuitive sense of object so in other words people use object and process in a way that might have a name space meaning for computer science but is actually less connected than we might want it to be to the formal math and this is pretty another last quote on this paper they write in as the separate chapters of a theory of concurrency all these different topics petri nets event structures and domains stone duality and process algebras these topics make concurrency something of a jigsaw puzzle one is then naturally led to ask whether the pieces of this puzzle can be arranged in some recognizable order and then they give a strong endorsement of the two-space framework so it's just interesting that these two spaces can be objects of relations that can be processes but also it can include context that's what this paper is about and algorithmic and game-like elements so if you can do it in golang if you can do it in the process algebra if you can use higher order reflexive process algebra maybe that's a two-space all right let's talk a little bit about ergodicity so this is going to have a few slides and definitions so on the top left a process is says to be ergotic if the ensemble average and time average are equivalent and here's a more formal definition from this is from the neurobytes site a more formal definition is from ole peters who's uh been at the santa fe institute an economics professor the ergotic hypothesis is a key analytical device of equilibrium statistical mechanics it underlies the assumption that the time average and the expectation value of an observable are the same where it is valid dynamical descriptions can often be replaced with much simpler probabilistic ones time is essentially eliminated from the model so this is in on the website neurobites it's a live and updating but here it's just static it's kind of thinking of two ways you can make purple from blue and red one is to have the spatial uh resolution increasing and pretty soon semantically it's interpreted as purple so if you doubt that you can zoom in on an old comic book and see how they made color which is by increasing the scale of resolution really small then it makes it look like a new color is there or you can flash things really fast and then it will appear as if there's purple and those equivalencies or ways of reaching and also deviations from ergodicity spatially and deviations from ergodicity temporally are what we're talking about now why does the ergodic assumption matter and what happens if this is empirically if this is violated so here's an example from modern portfolio theory which is what ole peters and others like glenn begerman study which is that and this is quote from neurobites modern portfolio theory stipulates there exists some optimal risk to return profile but the theory rests on the ergotic hypothesis by conflating an ensemble aggregate return with the individual's path-dependent return so it's really interesting um and this is related to the st petersburg paradox and a couple other economic ideas that basically the total population's performance or a hundred different portfolios performance at one snapshot is not the same thing as any portfolio being carried forward through time why does that matter how off on the wrong track would we be so here's some funny examples of ergodicity violations and also a hypothetical example of ergodicity so um number two from the neurobites if it takes one pregnant woman nine months to give birth nine pregnant women should only take one month and then this says recipe says bake for 60 minutes at 200 degrees but i'll just bake for 30 minutes at 400 degrees so like if if the degree minute is all things being equal why couldn't we just vaporize it with 10 000 degrees celsius for just a split second um it turns out that there's a process that has to play out and so yeah what do you think about this or how would you link it to anything else we talked about today um i'm like i don't know i think my brain is is uh i i i'm still giggling at the nine nine segment uh months i'll say something i i find those examples actually worse than just giving me no example because like ergodicity implies you have to have a system level demarcation and so you know to compare it to like nine pregnant women it's like well the the system is the woman so you can't really do that in the analytic it actually makes it harder to understand which is good good point yeah it's it's it's all right it's like but it just it yeah i think it makes it more confusing yeah i agree it's not a perfect mapping it's like syntactically it's the kind of error you'd be making like here's the ergotic in a lifestyle span but then if you go too down the rabbit hole we're trying to add the variables that's why we always pull back to the actual formalisms so here is from math encyclopedia and here it's i thought it was quite interesting so it's about trajectories of points and saying that as you'll hear with many other definitions that it's about visiting a phase space through space and that was sort of that first example here's what was interesting though when the birkhoff ergotic theorem had been proved it became clear that ergodicity is equivalent to metric transitivity measurement transitivity the more general situation it was no longer suitable to talk of the equality of time and space averages systems with an infinite so what kinds of systems are beyond spatial and temporal ergodicity systems with an infinite invariant or quasi-invariant measurement not only flows in cascades but also more general transformation groups and semi transformations groups categories semi groups we saw abelian semi groups earlier in the bar paper so it's almost like we're taking ergodicity even beyond the trajectory visitation idea i don't know where it's going in the way that the authors are using it but it's like there's there's like the you know the totally inadequate metaphor like of just making a total category error with the recipe and then there's the more empirically observable strategy violations and like you're going to have a bad strategy if you make if you're using a model with an ergotic assumption you're gonna do strategy that's just ridiculous you're gonna fit what seems to be the best or the most likely strategy and you're just gonna be way off base empirically but this is even taking it to another way okay let's just with a couple of these questions bring some of our clothes um i think you may have raised this one sarah but it was asking that this context by default generalizes the intrinsic contextuality of quantum theory to the case in which other properties of a context for example the order in which questions are asked also affect the distribution of a variable of interest this kind of direct influence is generally avoided by experiment design particularly by imposing no signaling conditions in physics but it's virtually inescapable when studying complex macroscopic systems so these are some pretty open questions how does context relate to erodicity especially when systems are changing their context or vice versa or both and how uh does this contextuality by specifically help us here and then how does fields in glazebrook build upon to maybe add another plank to this question okay here's another set of questions the existence of complementary observables has perplexed physicists for decades feignman characterized it as the only mystery of quantum theory how are such paradoxes of resolved at the implementation level if they are or worked around if they're not what happens when workaround mechanisms fail what is required to detect failure post hoc ability to prove this experimentally would begin to answer the broader question of how prior probabilities are represented in memory and how different sets of priors can be deployed in different contexts so how do we know what we don't know for a given context and how can a context be made to entail unknown unknowns how do we bound our not knowing and i think part of the way is going to be defining everything as the markov blanket which is going to be us internal states and blanket states is what we can observe and control and so everything is going to be conditioned on generative model of external states so we're never even going to peep outside of the blanket states we're going to be super serious about modeling ourself and the blanket states as interacting with and modeling external world states so instead of how tall is the tree you're going to have like you and the measuring device and the measurement in a system and maybe for some easy questions it's going to be total overkill but i think for other questions it's going to really open the door because once people frame it in this way what was an unknown unknown is going to be something that we can actually encapsulate so that's bringing us this question here's a quote we apply this development of ideas to recover among other things a new theoretical and inferential role for the markov blanket concept we read this in the introduction because it's a key point as the locus at which free energy is minimized and the epistemic knowledge barrier that keeps contextual information hidden from the observer so where is markov blanket coming into play what does this have to do with our modeling and learning and then just all these issues that we've been talking representation internal versus external perspectives and um where is this paper coming into play okay then sarah this is one of your suggestions slides from a couple weeks ago so but it's relevant here again so what maybe what if you want to anything that's interesting like about context or frame or reservoir just you're you know a high foraging philosopher of science so it's always good to hear what you think about it and my brain is pretty fried but i i think i can um my eyeballs are fried anyway um the thing that strikes me as a kind of contrast about what i'm able to read from this paper um is i i don't know i get the sense when they're writing in the paper that they just have this like you know how can we have the eye of god in like all contexts and shift like i you know or at least they're talking about that in the context of like modeling and talking about how our world actually is but um the thing that makes reservoir computing so powerful one of the things is is its non-ergodicity it's it's kind of natural partitioning it's it's um heterogeneity in some way um and so i don't know that's really interesting to me this kind of how can a body compute um and what are the attributes about a body that what are the kind of imperfections about the body that make it um that make it intelligent in that sense so so yeah you know in what way do channels form like these natural cutic natural joints or make these kind of boundaries and and why i don't know that's those are the interesting parallels for me yep that idea of like it's the the local consistency the organismality the ending of itself that's like the kantian organism concept but it's an aberration it's far from equilibrium from the global context which is just hydrogen molecules dispersed so it's like when you have a global disordering or some sort of perturbation and then you have a locally organized bubble it's then it's like it is our unique differences and situation that allow that agent to be adept in the niche it almost couldn't be another way or it's hard to see i don't know the thing that's interesting is like you don't even have to think about well i'm sure there is more you have to think about in terms of adaptation and you know response and things like this but you know how does a rock compute and and there was one paper that i was reading where they were talking about like well okay if you stick a rock in the oven you know it doesn't necessarily heat up in all the same places so you do have this kind of um you do have these these um non-homogeneous kind of attributes about the rock and and it's a it's a very loose analogy but that is essentially like this this um non-boltzmann type type distribution that is what makes these bodies computable in some way that yet still a little bit mysterious yep and it's like if it were um a little worm you said if it oh its body is set up with the muscle the connective tissue in the nerve so that it moves away from a prick from on the side it doesn't need to be computing where everything is if it's body is able to respond then it can do it needs to do in the niche okay but i mean you don't even need to be alive that's the thing that kind of blows my mind about how this computing works yes so implications next steps the first so this is good that they went there so the first question which is expanded upon more in this bestiava paper is exploring contextuality for human measurements cool a second is understanding how context switching is implemented at the neurocognitive level so that's the feigning quote that we just looked into but basically what is happening when it's like somebody's your friend you're in a narrative of friendship and i get from the cultural down to the you know the day down to the second down to the micro expression you're in a narrative and then a stimuli happens and a context switches and that trajectory is still a unitary your relationship with them is still a but it's changed in a way that the context has changed like how does that get implemented this combination of the high precision priors and the high precision inputs how do we get the best of both worlds with this rapid attentional modulation as brains into and away from objects and then context switching those are features that are just not the scope of shannon disinfo theory and they're much more semantic questions a third and broader question is whether intrinsic contextuality can be detected and characterized at multiple scales in complex systems generally by formulating a scale-free model of contextuality in category theory terms and in particular by employing the concepts of classifiers and information we hope to have opened a new pathway towards addressing these questions in multiple ways or addressing these questions in multiple systems so maybe if there is something special to be said about contextuality or the violations thereof two sides of the same coin maybe we could identify in a less biased way what scales of organization are relevant or just detect new dynamics of complex systems because there's a lot of recent work showing that like with complex systems they can emit structured outputs that have the signature of white noise or have a signature of nothing like an encrypted file is going to have a distribution syntactically that's going to be looking like nothing different from the frequentists expectations but it's a semantically encoded local logic that is allowing communication through that channel so if all you have is the white noise detector you're going to think up that looks like the signal is not containing information but actually semantically it is so there's there could be systems just in and around us that are doing things we don't even see and that would be no weirder than if there was wavelengths we didn't see we know that's true we know that there's things that are too big or too small for us to see so could there be things informationally or from a narrative perspective that we're just not perceiving doesn't seem out of the picture so i think it's a it actually is a major area and you know for those who have participated or listened to this deep in chris fields was the one who recommended this paper to us and just said i think this is a topic that is not really thought about in the active inference community that much but it will become important later i don't even know what topic he was referring to but this was the paper that really raised some huge questions so all kinds of questions that we have i think it's fair enough just to say um thanks to r2 3 brave and awesome participants thanks sarah and blue because that was a fun convo and we definitely laughed and memed because it was like the hardest paper i think we've read yet definitely but you know the paper was so well written like i am not a big fan of reading like math papers and lemma this and theorem that like i'm not i usually don't enjoy it at all but i really did find this paper to be very well written and easy to follow i agree when there was a formalism it's a theorem definition figure formalism is like okay i'm not going to understand that but i know that that's just the theorem so i could just say okay got it something i don't understand put it away encapsulate that so i agree um any last thoughts sarah blue i just thought i was almost wishing like as i was reading the paper like for some concrete examples but we would have ended up with like a 65 page paper so i guess that's our job to be the concrete example you know yep and that will be fun to talk to chris about so um let's um until the time we actually get to talk about this paper which is going to be on march 9th so between now and march 9th let's just chill you know do our other jobs instead of just reading this paper think about what would be really powerful and meaningful to ask them